PRACTICE WITH RATES

1. The volume of a tree is given by $V = \frac{1}{12\pi}C^2 h$ where C is the circumference of the tree in meters at ground level and h is the height of the tree in meters. Both C and h are functions of time t in years.

A. Find a formula for $\frac{dV}{dt}$. What does it represent in practical terms?

$$\frac{dV}{dt} = \frac{d}{dt}\left(\frac{1}{12\pi}C^2 h\right) = \frac{1}{12\pi}\frac{d}{dt}(C^2 h) = \frac{1}{12\pi}\left[\frac{d}{dt}(C^2)h + C^2\frac{d}{dt}h\right] = \frac{1}{12\pi}\left[2C\frac{dC}{dt}h + C^2\frac{dh}{dt}\right] = \boxed{\frac{Ch}{6\pi}\frac{dC}{dt} + \frac{C^2}{12\pi}\frac{dh}{dt}}.$$

This represents how the volume of a tree changes as time increases.

B. Suppose the circumference grows at a rate of 0.2 meters/year and the height grows at a rate of 4 meters/year. How fast is the volume of the tree growing when the circumference is 5 meters and the height is 22 meters?

Assume : $\frac{dC}{dt} = \frac{1}{5}$ and $\frac{dh}{dt} = 4$

Given: $C = 5$, $h = 22$

$$\frac{dV}{dt} = \frac{Ch}{6\pi}\frac{dC}{dt} + \frac{C^2}{12\pi}\frac{dh}{dt} = \frac{5(22)}{6\pi}\left(\frac{1}{5}\right) + \frac{5^2}{12\pi}(4) = \frac{22}{6\pi} + \frac{25}{3\pi} = \frac{1}{3\pi}(11+25) = \frac{36}{3\pi} = \boxed{\frac{12}{\pi}}$$

2. A. When the radius of a spherical balloon is 10 cm, how fast is the volume of the balloon changing with respect to change in its radius?

Volume of a sphere of radius R: $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dr} = \frac{d}{dr}\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi\frac{d}{dr}(r^3) = \frac{4}{3}\pi(3r^2) = 4\pi r^2$$

$$dV/dr\Big|_{r=10} = 4\pi(10)^2 = \boxed{400\pi}$$

B. If the radius of the balloon is increasing by 0.5 cm/sec, at what rate is the air being blown into the balloon when the radius is 6 cm?

$$\frac{dV}{dt} = \frac{d}{dt}\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi\frac{d}{dt}r^3 = \frac{4}{3}\pi(3r^2)\frac{dr}{dt} = 4\pi r^2\frac{dr}{dt}$$

Given $dr/dt = 1/2$ $r = 6$, the volume of the balloon is changing at a rate of

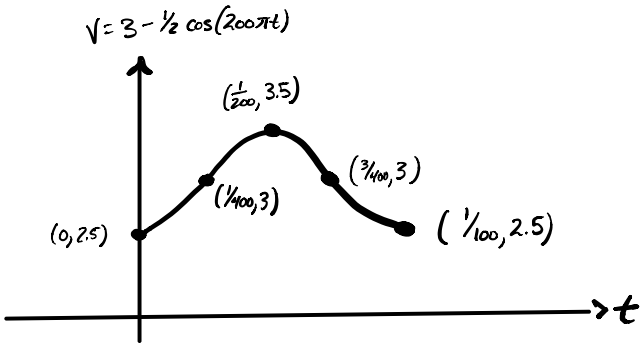
$$\frac{dV}{dt} = 4\pi(6)^2\left(\frac{1}{2}\right) = 2(36)\pi = \boxed{72\pi}$$

C. When the volume of the balloon is 50 cubic cm, at what rate is the radius of the balloon changing?

There's information missing from this problem.

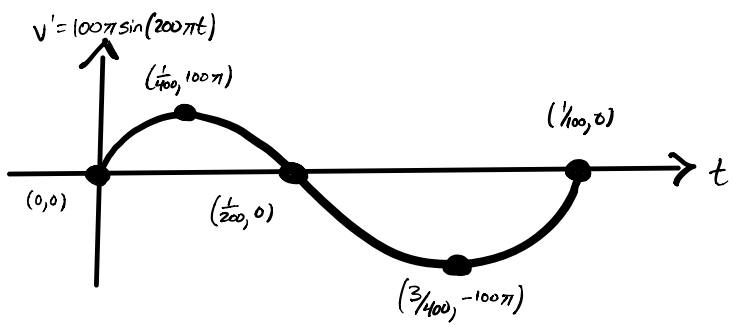
3. When hyperventilating, a person breathes in and out very rapidly. A spirogram is a machine that draws a graph of the volume of air in a person's lungs as a function of time. During hyperventilation, the person's spirogram trace might be represented by $V = 3 - 0.05\cos(200\pi t)$ where V is the volume of air in liters in the lungs at time t minutes.

A. Sketch a graph of one period of this function.



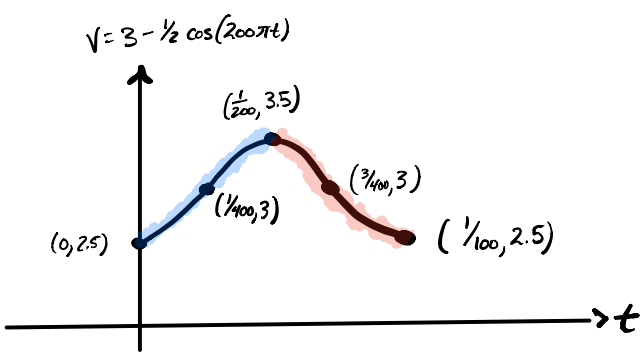
B. What is the rate of flow of air in liters/minute? Sketch a graph of this function.

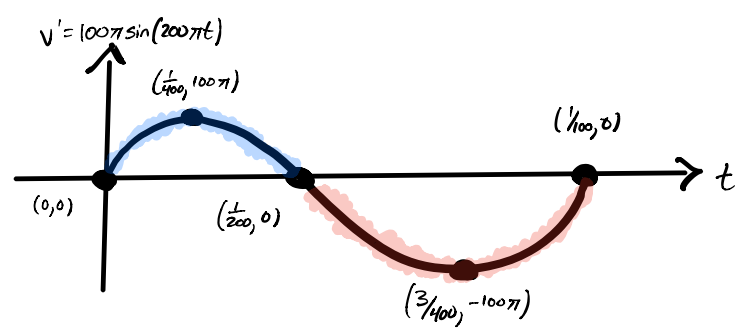
$$\frac{dV}{dt} = \frac{d}{dt}\left(3 - \frac{1}{2}\cos(200\pi t)\right) = 0 - \frac{1}{2}(-\sin(200\pi t))\,200\pi = 100\pi\sin(200\pi t)$$



C. Mark the following on each of the graphs above.

i) the interval when the person is breathing in
ii) the interval when the person is breathing out
iii) the time when the rate of flow of air is a maximum when the person is breathing in





(i) & (ii) Breathing in is marked in blue (volume increasing), breathing out in red (Volume decreasing).

(iii) The rate of flow is maximal while breathing in at $t = 1/400$.